Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/06/2015
PI Name:	Marr, David W.M. Ph.D.		
Project Title:	Fabrication, Crystallization, and Folding of Complex Colloidal Molecules under the Influence of Applied External Fields		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dmarr@mines.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-273-3720
Organization Name:	Colorado School of Mines		
PI Address 1:	Chemical and Biological Engineering		
PI Address 2:	1500 Illinois Street		
PI Web Page:			
City:	Golden	**State:**	CO
Zip Code:	80401-1887	**Congressional District:**	7
Comments:	https://		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
Contact Email:	william.v.meyer@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Ning (Colorado School of Mines) Solomon, Michael (University of Michigan, Ann Arbor)		
Grant/Contract No.:	NNX13AQ54G		
Performance Goal No.:			
Performance Goal Text:			

Colloidal particles are technologically important materials with applications ranging from consumer products to plasmonic metamaterials to drug delivery agents. With phase behaviors and assembly characteristics that closely resemble atoms and molecules, they are also excellent molecular analogues that can grow into functional materials with high efficiency. The range of colloidal structures and molecular mimics we can obtain, however, is currently limited and we remain far from truly accessing the diversity of phases desired for both deeper fundamental understanding and associated technological application. This is due primarily to limited control of the intrinsic colloidal interactions in conventionally isotropic building blocks and a reliance on non-directed assembly.
To address these issues, we propose the use of anisotropic external fields including optical and hydrodynamic fields to create anisotropic colloidal building blocks (primarily on ground). We also propose to use anisotropic electric fields to

Task Description:	direct their assembly into complex structures (primarily in space). This combination of rational design of colloidal composition and architecture and versatile control of the range, strength, and directionality of interparticle interactions will allow the study of two outstanding scientific questions particularly appropriate for investigation at the International Space Station: (i) How does complex symmetry evolve during the crystallization of anisotropic colloidal molecules, such as achiral dimer particles, racemic mixtures of chiral tetramers, and pure enantiomeric tetramers? (ii) How will the collective fluctuations of nodes (tetramers) and linkers (colloidal chains) induce the formation of mixed networks with non-linear mechanical properties? Results from these investigations will lead to better control of the non-covalent assembly of molecules and scalable processing of advanced materials with tailored structures and properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In-situ manipulation of anisotropic interactions and dynamic pathways, based on rational colloidal particle design and proper use of external fields, could lead to crystalline and aperiodic structures beyond those seen in nature. With the wide variety of attributes that characterize anisotropic interactions, we will probe previously inaccessible regions of complex phase space. Our studies will move significantly beyond hard spheres and towards experimental models appropriate for studying fundamental questions associated with complex symmetries. For example, how does chirality of the assembled structure evolve during the nucleation and crystallization of achiral molecules, chiral molecules in racemic mixtures, or in pure enantiomers? Furthermore, our models of giant colloidal chains will allow for the manipulation and observation of the effect of site-specific interactions on chain dynamics at experimentally accessible length and time scales. Models that can be easily visualized like these will eventually lead to better understanding of complex phenomena such as polymer chain dynamics, chain reptation, interpenetrating double network dynamics, and protein folding. Results from this study will lead to improved control of non-covalent assembly of molecules, efficient tailoring of lattice symmetries, and the scalable processing of nano-structured materials. The development of new colloidal molecules and their associated assembled structures may have considerable technological impact as well. Anisotropic interactions developed in this proposal, both at the particle level and via applied fields, have been shown generally to lead to arrays with reduced symmetry and enhanced directionality. Such structures interact with a broad range of electromagnetic radiation in unique ways and can exhibit collective photonic, plasmonic, mechanical, electronic, or magnetic properties that are not manifested at the level of single particles. As a result, they have significant potential as next-generation functional materials. For example, dielectric arrays with diamond-like or quasicrystalline lattices have been proposed as ideal photonic crystals that can manipulate light propagation in three dimensions, forming the basis for next-generation all-optical communication technologies. Chiral tetramers studied here could exhibit distinct plasmonic properties resulting from the collective coupling between particles in the pyramid, as confirmed in numerical modeling, which have great potential for electromagnetic metamaterials and high-resolution sensors. The understanding of mirror-symmetry breaking in racemates is also important for developing improved strategies for separation of pharmaceutical molecules. In addition, anisotropic particles can be used to generate new kinds of colloidal networks with well-defined coordination and connectivity. By combining families of tetramers and chains, double networks – two distinct, but interpenetrating, networks of nodes and linkers – could be formed with combined high mechanical stiffness and high toughness. Colloidal networks are very common in a range of pharmaceutical and consumer production applications because of their ability to impart a yield stress that can stabilize active components uniformly throughout the formulation. Developing methods to produce colloidal networks with defined coordination, connectivity, and topology would expand the use and performance of colloidal stabilizing networks in these materials. Ultimately, successful implementation of our proposed research will provide the fundamental knowledge necessary for the development of technologies to design and control matter with tailored structures and properties.
Task Progress:	Our progress during this period includes: 1) Our experiments have shown that there are interparticle repulsions within our linear optical trap. To overcome these, we have introduced a magnetic field into our system and used this to form colloidal chains with specific lengths. Colloid chain length is determined via a balance of the hydrodynamic shear force, the applied magnetic field strength, and the optical force. 2) We further fixed the assembled colloidal chains in situ by using thiol-functionalized magnetic beads and a functionalized polyethylene glycol crosslinker. With the help of Michael-addition chemistry, a fast and robust linkage between individual colloidal particles has been achieved during assembly. We have successfully prepared short chains with different flexibilities, all tunable due to linkers of different lengths. 3) We have also fabricated colloidal chains in bulk using the same cross-linking chemistry. With this approach, very long (~mm) and stable (tested up to 30 days) chains were prepared by optimizing both reaction temperature and time. 4) In addition to the study of binding kinetics, we have initiated investigation of lock-and-key colloid incorporation into our chain architecture. From concentration measurements of lock, key, and lock-key specifically bound pairs we have extracted equilibrium constants and free energies of formation of lock-key pairs at different salt and polymer conditions. We are furthermore performing equilibrium self-assembly studies of lock-key colloids in the presence of electrolyte and depleting polymer to ascertain the effect that the degree of shape complementarity has on the formation of specific bonds and colloidal molecules.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Wu N, Yang T, Ma F, Marr DWM. "Fabrication and Assembly of Colloidal Molecules Using External Fields." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014